

YOUXEL E-Services Module

Belal Bahr

Product Manager

27th of Dec, 2025

Executive Summary

The E-services Module is a strategic digital initiative designed to transform YOUXEL's service delivery from a fragmented, manual process into a centralized, "data-first" digital hub. Currently, users face significant friction, relying on phone calls, physical visits, and emails to interact with IT, HR, and Facilities. This module eliminates these bottlenecks by consolidating service discovery, submission, payment, and tracking into a single intuitive interface.

Product Objectives

- **Centralization:** Provide a unified searchable directory for all organizational services.
- **Operational Efficiency:** Reduce overhead by automating workflows, approvals, and SLA monitoring.
- **Transparency:** Empower users with real-time status tracking and automated notifications, reducing support inquiries.

Technical & Strategic Approach

The system is built on a modular monolith architecture using Node.js (NestJS) and React 18, ensuring a balance between rapid MVP deployment and future microservices scalability.

MVP Strategy

Using the RICE scoring methodology, the product prioritizes high-reach, high-impact features such as the Service Catalog and Status Tracking Dashboard. The MVP focuses on validating the core hypothesis: *Will users adopt digital self-service over manual methods?* To ensure speed to market, advanced features like dynamic forms and native mobile apps are deferred in favor of a robust, responsive web experience.

Business Case

Current Problem: The organization processes 2,000 manual support requests per week, consuming approximately 400 staff hours across HR, IT, and Facilities.

Digital Solution: Transitioning to a self-service model is projected to reduce manual handling by 75%, bringing the operational load down to 100 hours per week.

Financial Impact: Based on a standard internal rate, this efficiency gain results in \$1.2M in annual operational savings.

ROI: With an estimated investment of \$180K for 6 person-months of development, the project achieves a full payback in less than 2 months.

1. Module Overview and Objectives

1.1 Purpose of Module

The E-services module serves as a centralized digital service hub. Instead of users navigating different departments or offline forms, this module consolidates all interactions into a single, intuitive interface.

Core functionalities include: service request submission and tracking, document management, integrated secure payments, and an automated support desk.

This initiative accelerates YOXEL’s digital transformation by reducing operational overhead, eliminating paper-based bottlenecks, and creating a "data-first" culture where every user interaction is measurable.

1.2 Target Users

User Group	Need	Pain	Expectation
System Users (External Customers)	Submit service requests, make payments, track applications	Current process requires phone calls, emails, physical visits	Mobile-first experience, instant confirmation, transparent status
Service Providers (Internal Staff)	Centralized service request interface for HR, IT, facilities	Fragmented systems, manual tracking, lost requests	Quick submission, approval routing, request history
Administrators	Request processing workflow, user management, reporting	Manual triage, no visibility into bottlenecks	Configurable workflows, SLA monitoring, bulk actions

Note: These personas are hypothesis-driven; validation planned through interviews, surveys, and focus groups during discovery sprint.

2. Feature Definition and User Stories

2.1 Key Features

1. A unified searchable directory (catalog) of all available services.
2. Dynamic forms that change based on user input.
3. Integrated payment gateways that support credit cards and digital wallets.
4. A status tracking dashboard.
5. Automated notifications (SMS, email, push).
6. Documents upload and verification.

2.2 User Stories

Feature 1:

As an end customer, I want to browse available services, understand requirements, and submit a request with required documents so that I can initiate a service without visiting physical offices.

Acceptance Criteria:

- User views categorized service catalog with search/filter
- Each service displays: description, required documents, estimated processing time, fees
- Form validation prevents submission with missing required fields
- User receives confirmation with tracking ID immediately after submission
- System stores submission timestamp and initial status

Feature 2:

As an admin user, I want to define approval stages, assign staff roles, and set SLA thresholds so that services route correctly and performance is measurable.

Acceptance Criteria:

- Admin creates multi-stage workflows (e.g., L1 Review, L2 Approval, Payment, Fulfillment)
- Each stage assigned to user role with escalation rules
- SLA timers configurable per stage (e.g., "L1 Review must complete within 24 hours")
- System auto-escalates overdue tasks to supervisors
- Workflow changes apply to new requests only (existing requests follow original workflow)

Feature 3:

As an end customer, I want to pay service fees securely through multiple payment methods

so that I can complete my service request without cash/bank visits.

Acceptance Criteria:

- System displays fee breakdown before payment
- Supports credit/debit cards, bank transfer, mobile wallets
- PCI-DSS compliant payment flow (tokenization, no storage of card data)
- User receives digital receipt via email and in-app
- Failed payments trigger retry mechanism with clear error messaging
- Refund workflow exists for cancelled/rejected requests

Feature 4 (and 5):

As an end customer, I want to view current status and history of my service requests so that I know what's happening without contacting support.

Acceptance Criteria:

- System Dashboard shows all requests with current stage (Submitted, Under Review, Approved/Rejected, Completed)
- Each status change displays timestamp and responsible party
- System sends notifications on status changes (SMS/email + push if mobile)
- User can filter/sort by date, status, service type
- Rejected requests show reason and remediation steps

Feature 6:

As an operations staff member, I want to review uploaded documents, request corrections, and approve/reject them so that only valid requests proceed to fulfillment.

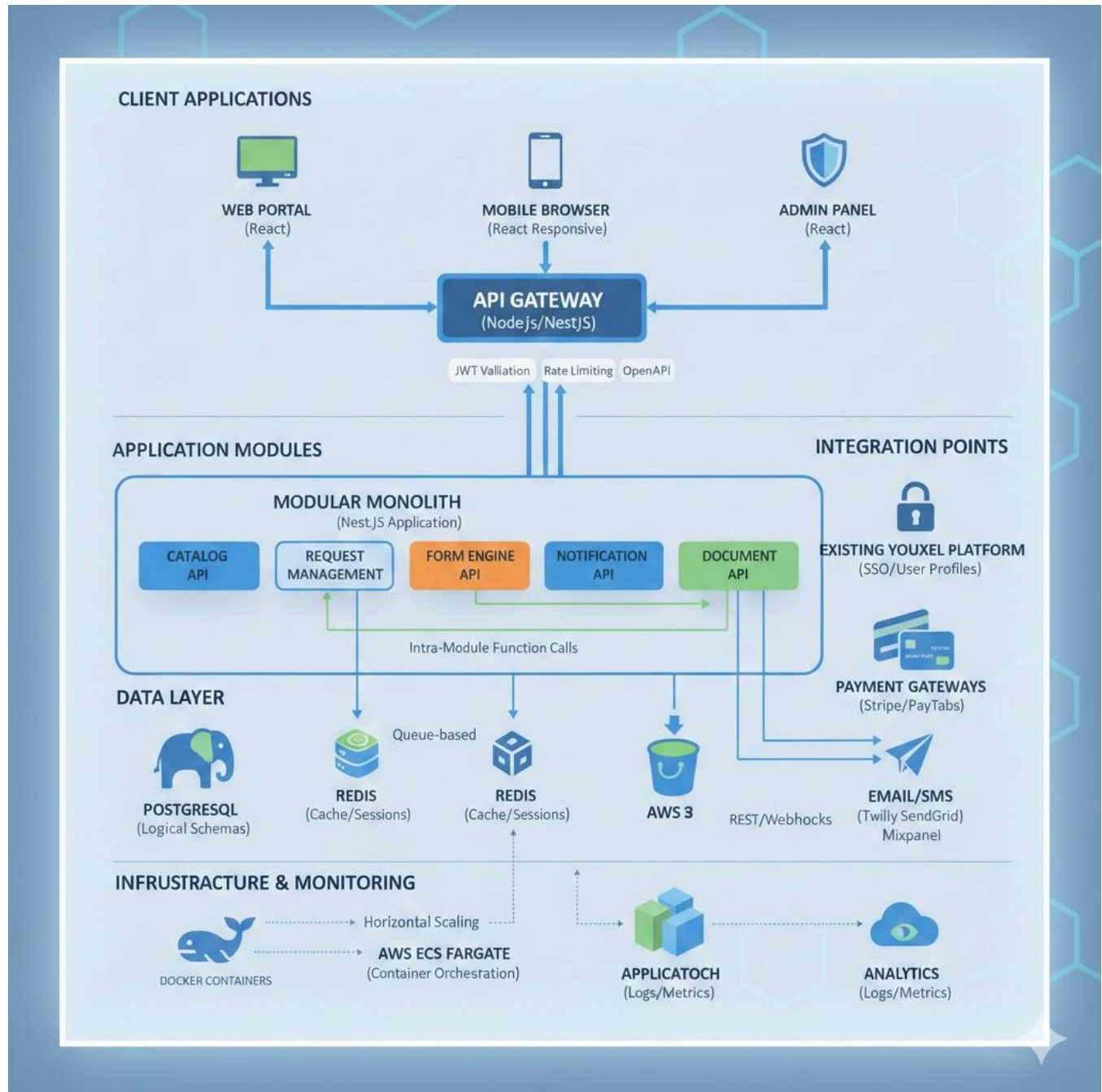
Acceptance Criteria:

- Staff views queue of pending document reviews sorted by submission date
- Each document displays in-app viewer (PDF, images supported)
- Staff can mark document as Approved, Rejected (with reason), or Request Resubmission
- Customer receives notification if resubmission needed with specific feedback
- System logs all verification actions with staff ID and timestamp
- Bulk actions available for processing multiple similar documents

3. System Architecture and Technical Specifications

3.1 Architecture Overview

The module uses a modular monolith architecture for MVP, with clear service boundaries that enable future extraction into independent microservices as load requires.



System Architecture Diagram

Frontend:

- React 18 for web portal, admin panel, and public website
- Responsive design for mobile browsers (will defer React native app)

Backend:

- Node.js with NestJS framework
- API Gateway pattern: single entry point, JWT validation, rate limiting
- RESTful APIs with OpenAPI
- Modular structure (can run as modules in monolith for MVP):
 - Catalog API
 - Form Engine API
 - Request Management API
 - Payment API
 - Notification API
 - Document API

Data:

- PostgreSQL single instance, logical schemas per service
- Redis (sessions and cache)
- AWS S3 for documents storage

Infrastructure:

- Docker containers
- AWS ECS Fargate
- Application Load Balancer
- CloudWatch for logging and monitoring

Intra-monolith direct function calls for module communication. Database access via logical schemas; modules query only their own schema.

Integration Points:

- Existing YOXEL platform: SSO, user profiles, notification service
- Payment gateways: Paymob /Fawry (dual integration for redundancy)
- Email/SMS: Twilio SendGrid
- Analytics: Google Analytics, Mixpanel, internal BI tools

3.2 Technical Requirements

1. Scalability:

- Client-side rendering for instant field updates, server-side validation for security.
- Horizontal scaling for API services (AWS ECS)
- Catalog search: Elasticsearch or PostgreSQL full-text search with trigram indexes

- Queue-based (AWS SQS) payment processing for resilience, max 5-second processing time
- Target load: 10,000 concurrent users, 8k requests/month at launch

2. Security:

- Payment data never stored on YOXEL servers, tokenization via gateway, compliance with standards and policies.
- Role-based access control (RBAC) with attribute-based rules.
- Data encryption at rest and in transit.
- File uploads virus scanning, mime-type validation, size limits (5MB/file), sandboxed storage.
- API security JWT with 15-min expiry + refresh tokens.
- API rate limiting per user/IP.
- Admin actions require MFA for form publishing, audit log for all config changes.
- Regular penetration testing and vulnerability scanning.

3. Data Privacy:

- GDPR/local data protection law compliance.
- Users have the ability to download their data and delete their accounts.
- Data retention policies and user consent management for data processing.
- Access logging provides a compliance trail by recording who accessed each request and when.

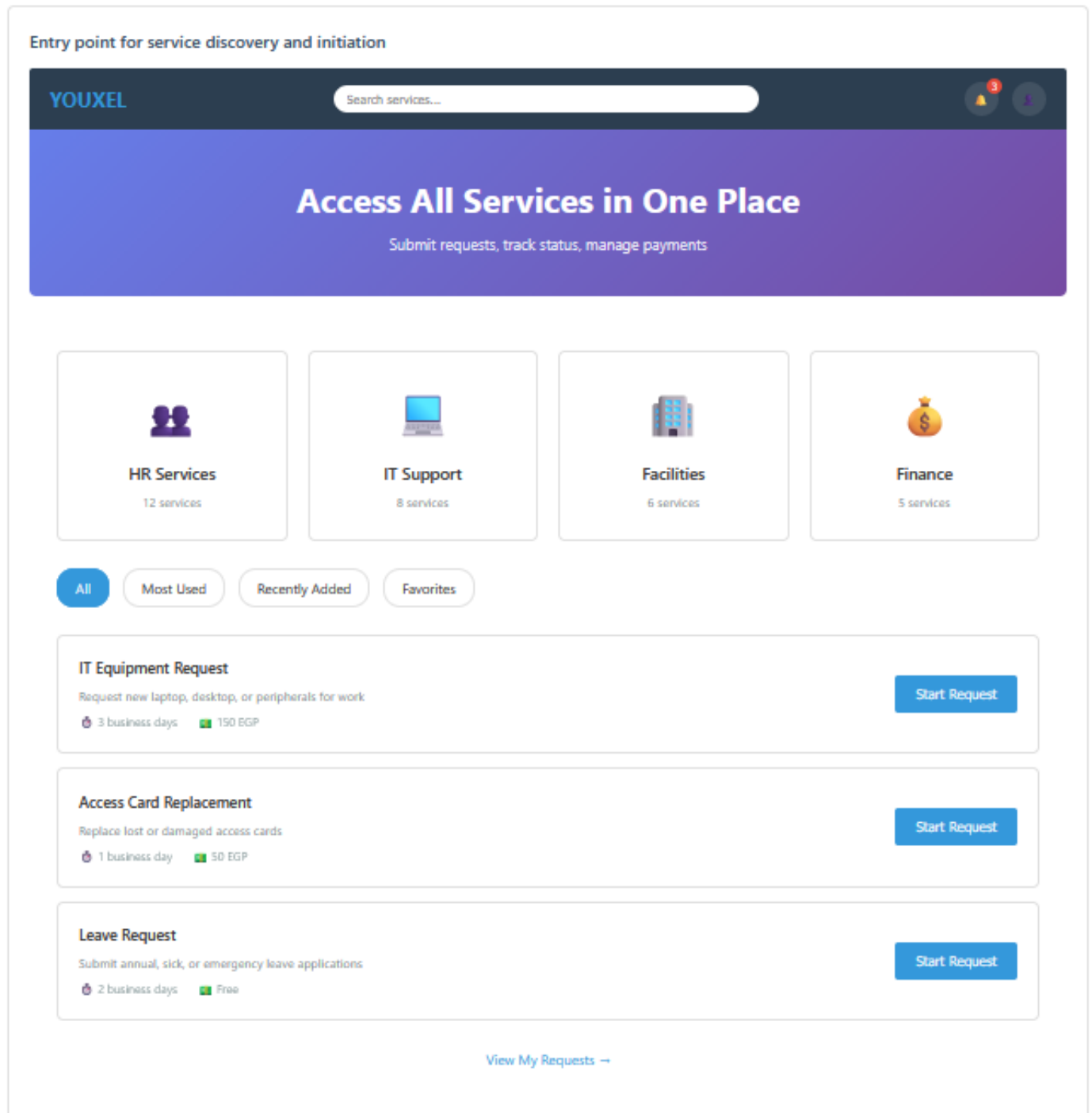
4. Integration Standards:

- Payment webhooks implement idempotent processing by deduplicating on transaction ID, enforce a 30-second timeout, and retry up to three times using exponential backoff.
- Notification delivery guarantees at-least-once delivery for critical events, while informational notifications are delivered on a best-effort basis.
- The search API supports filtering via query parameters and uses cursor-based pagination
- Structured logging (JSON format for centralized aggregation).

4. Design Mockups and Interaction Flows

4.1 Mockups

Screen 1: Service Catalog (Landing Page)



Screen 2: Service Request Form

Step 1 of 3: Request Details

1

Request Details

2

Documents

3

Review & Pay

IT Equipment Request • Estimated processing: 3 business days • Fee: 150 EGP

Basic Information

Equipment Type *

Select equipment type

Justification *

Explain why you need this equipment...

0 / 500 characters

Urgency *

☒ Standard

☐ Urgent (+50 EGP)

Urgent requests are processed within 1 business day

Delivery Details

Delivery Location *

Cairo Main Office

Preferred Delivery Date *

mm / dd / yyyy

Minimum 5 business days from today

Required Documents

✓ Manager Approval Email

Forwarded email from your direct manager

✓ Cost Center Code

Department budget allocation document

Need Help?

What if I don't have manager approval? ▼

Can I request multiple items? ▼

Contact Support

Save as Draft

Next: Upload Documents →

Screen 3: Document Upload

Step 2 of 3: Upload Required Documents



IT Equipment Request • Estimated processing: 3 business days • Fee: 150 EGP

Document 1: Manager Approval Email Required

 manager_approval_20241227.pdf
2.3 MB

✓ Verified ✕

Document 2: Cost Center Code Required



Drag and drop file here or click to browse
Accepted: PDF, JPEG, PNG (max 5MB)

Document 3: Additional Justification Optional



Drag and drop file here or click to browse
Accepted: PDF, JPEG, PNG (max 5MB)

← Back

Next: Review & Submit →

⚠ Please upload all required documents before proceeding

Screen 4: Review & Payment

Step 3 of 3: Review Your Request and Complete Payment

✓

Request Details

✓

Documents

3

Review & Pay

Request Details

Edit

Service:

IT Equipment Request

Equipment Type:

Laptop

Urgency:

Standard

Delivery Location:

Cairo Main Office

Preferred Date:

January 5, 2025

Uploaded Documents

Edit

✓ Manager Approval Email (manager_approval_20241227.pdf)

✓ Cost Center Code (cost_center_IT2024.pdf)

Payment Summary

Service Fee:

150 EGP

Urgency Fee:

0 EGP

Total:

150 EGP

Select Payment Method

Credit/Debit Card

VISA Mastercard

Mobile Wallet

Fawry

Bank Transfer

Manual verification required

Card Number

1234 5678 9012 3456

Expiry Date

MM/YY

CVV

123

🔒 Payment secured by Stripe

☐ I agree to the [terms and conditions](#)

← Back

Submit Request & Pay 150 EGP



Your request has been submitted successfully!

REQ-2024-001234

Service:	IT Equipment Request
Submission Date:	Dec 27, 2024, 2:45 PM
Estimated Completion:	Jan 3, 2025
Status:	Under Review

✓

2

3

4

What happens next?

Request Submitted

Dec 27, 2024, 2:45 PM

Document Verification

In progress

Due by Dec 28, 2024, 5:00 PM

Manager Approval

Pending

Fulfillment

Pending

View Request Details

Screen 6: Request Tracking Dashboard

My Requests

All Statuses

All Services

Last 30 days

Search by request ID or service name...

Export CSV

3

Active Requests

12

Completed

1

Pending Payment

4.2

Avg. Processing Time (days)

Request ID	Service	Submission Date	Status	Actions
REQ-2024-001234	IT Equipment Request	Dec 27, 2024	Under Review	
REQ-2024-001198	Access Card Replacement	Dec 20, 2024	Completed	
REQ-2024-001156	Leave Request	Dec 15, 2024	Approved	
REQ-2024-001089	Expense Reimbursement	Dec 10, 2024	Rejected	

Showing 1-10 of 47 requests

←

1

2

3

...

5

→

Screen 7: Request Detail View

REQ-2024-001234

IT Equipment Request

Under Review

Status Timeline

✓ Request Submitted

Dec 27, 2024, 2:45 PM

Submitted by: You

✓ Payment Confirmed

Dec 27, 2024, 2:46 PM

System: Payment processed successfully

📄 Document Verification

In progress

Assigned to: Sarah Ahmed (L1 Reviewer)

🕒

Due by Dec 28, 2024, 5:00 PM (18 hours remaining)

4 Manager Approval

Pending

5 Fulfillment

Pending

Details

Communication

Payment

Request Information

Equipment Type:

Laptop

Urgency:

Standard

Delivery Location:

Cairo Main Office

Preferred Date:

January 5, 2025

Justification:

Current laptop experiencing hardware failures, impacting daily work productivity. Need replacement to maintain workflow.

Uploaded Documents

✓ manager_approval_20241227.pdf (2.3 MB)

✓ cost_center_IT2024.pdf (1.8 MB)

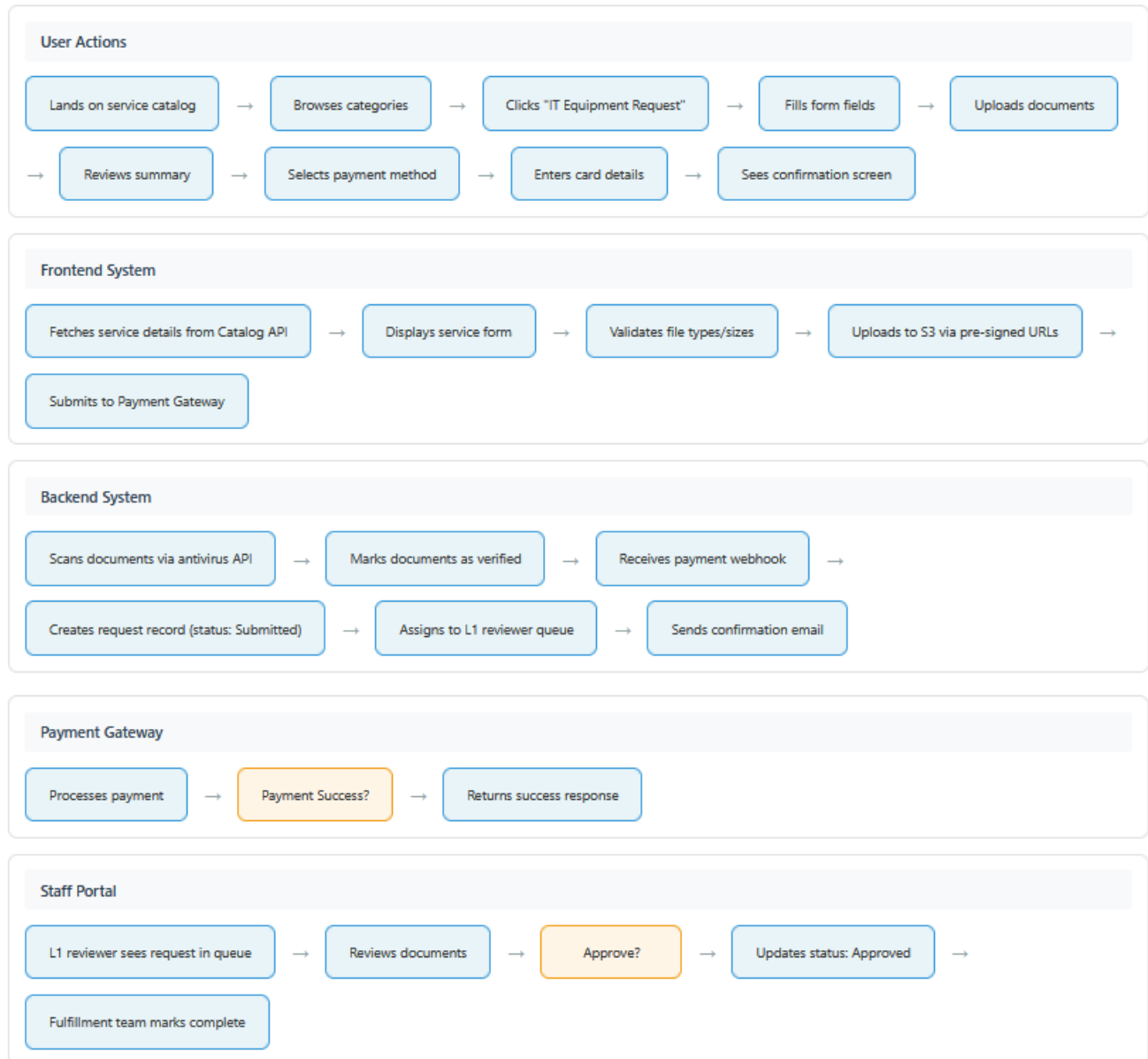
Cancel Request

Contact Support

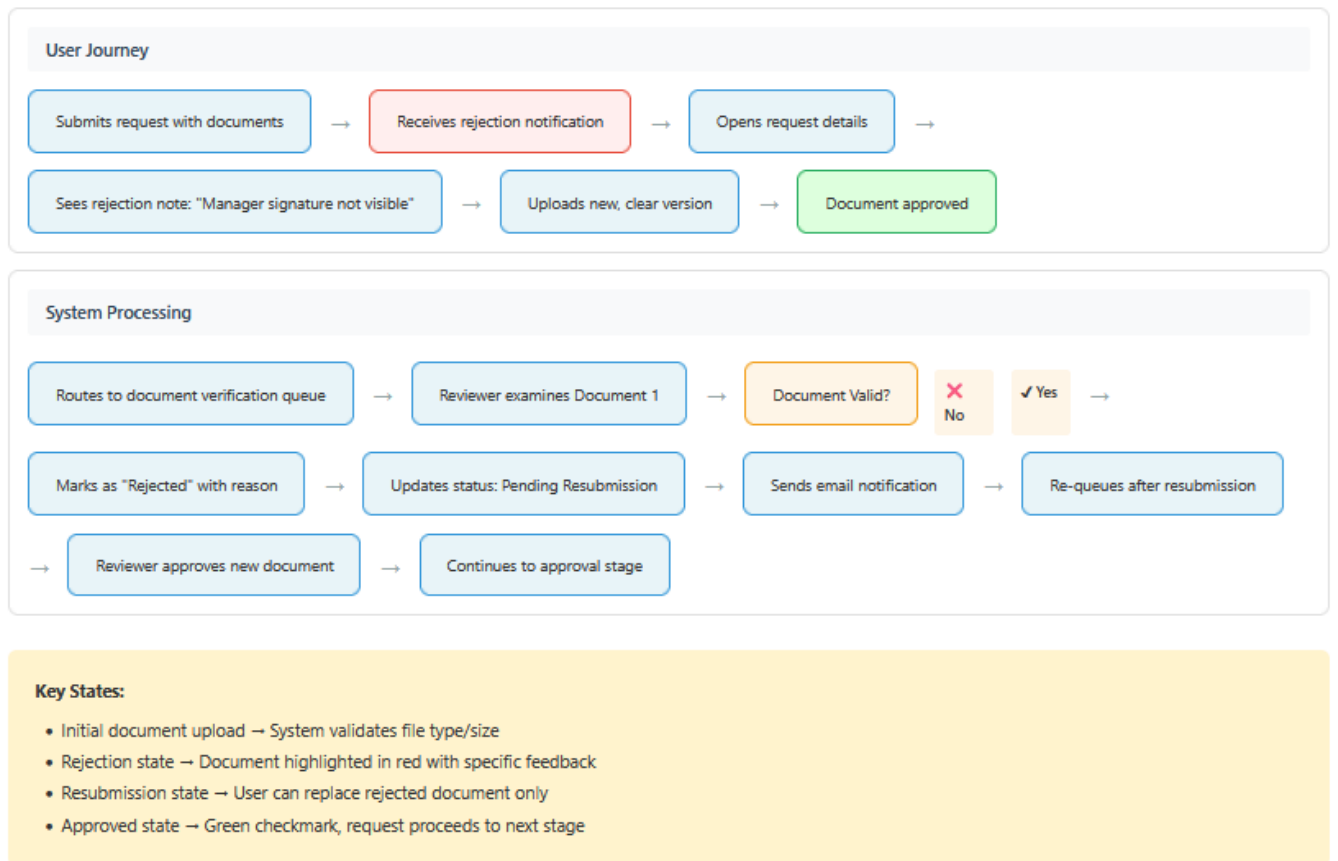
14

4.2 Interaction Flows

Interaction Flow 1: Happy Path - End-to-End Request Submission

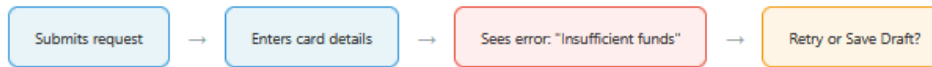


Interaction Flow 2: Document Rejection & Resubmission Loop



Interaction Flow 3: Payment Failure & Retry

User Actions



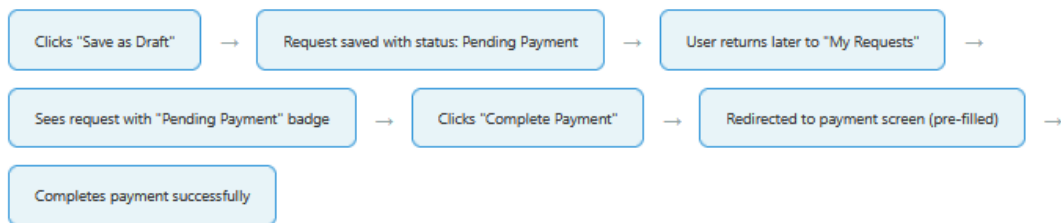
Option 1: Try Different Card

Option 2: Save as Draft

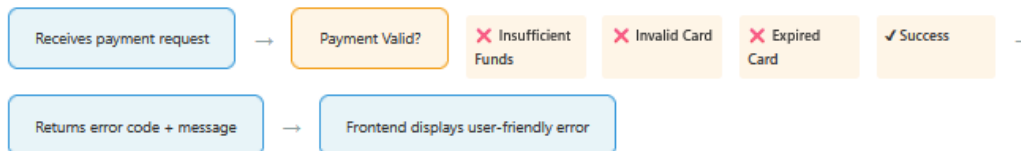
Option 1: Immediate Retry



Option 2: Save Draft Flow



Payment Gateway Processing



Error Handling Strategy:

- **Payment failure** → User stays on payment screen, form data preserved
- **Clear error messaging** → "Insufficient funds" not "Error code 2043"
- **Retry mechanism** → Allow immediate retry with different card
- **Draft save option** → User doesn't lose progress if unable to pay immediately
- **Status tracking** → "Pending Payment" visible in dashboard with "Complete Payment" CTA

5. Prioritization and MVP Definition

5.1 Prioritization

Methodology: **RICE Scoring**

RICE evaluates each feature on four dimensions:

- Reach: Number of users/requests affected per time period
- Impact: Degree of value delivered (scale: 0.25 = minimal, 0.5 = low, 1 = medium, 2 = high, 3 = massive)
- Confidence: Certainty in estimates (scale: 50% = low, 80% = medium, 100% = high)
- Effort: Person-months required for implementation

Feature	Reach (users/quarter)	Impact (0.25-3)	Confidence (50-100%)	Effort (person-months)	RICE Score	Priority
1. Unified Searchable Service Catalog	5,000	3	100%	2	7,500	1st
4. Status Tracking Dashboard	5,000	2	100%	1.5	6,667	2nd
3. Integrated Payment Gateways	3,500	2	80%	3	1,867	3rd
6. Document Upload and Verification	4,000	1	100%	2.5	1,600	4th
5. Automated Notifications	5,000	1	100%	1.5	3,333	5th
2. Dynamic Forms	3,000	0.5	80%	3	400	6th

The catalog and tracking dashboard scored highest because they reach every user on every interaction. The service catalog is the entry point, without it, users cannot discover or initiate requests. The tracking dashboard addresses the primary pain point driving users to the platform: "Where is my request?" Both have massive impact (3 and 2 respectively) because they eliminate the current phone/email process entirely, and both are relatively low-effort compared to payment integration.

Dynamic forms ranked last because reach is limited to services requiring conditional logic, impact is low (nice-to-have UX improvement, not a blocker), and effort is high due to form builder complexity, validation engine, and conditional rendering logic. Static forms deliver 80% of the value at 20% of the cost.

Payment and dynamic forms received 80% confidence (not 100%) because we have less certainty about integration complexity and user adoption of advanced features. All others received 100% confidence as they're standard CRUD operations with well-understood implementation patterns.

5.2 Minimum Viable Product (MVP)

The MVP tests one critical assumption: Will users choose digital self-service over the existing phone/email process?

MVP delivers the minimum feature set to eliminate this entire flow: Catalog lets users discover services and requirements without calling. Payment integration lets users pay immediately without bank visits. Status dashboard eliminates "where is my request?" calls. Document upload eliminates physical visits.

Advanced features (dynamic forms, multi-gateway payments, SMS notifications, workflow automation) improve the experience but don't change the fundamental value proposition: submit and track requests without human interaction. These optimizations only make sense after validating users will adopt the base product.

MVP Scope:

- **Feature 1: Unified Searchable Service Catalog**
Read-only catalog displaying 10 initial services (5 IT, 3 HR, 2 Facilities). Basic text search (no filters). Each service shows: name, description, processing time, fee, required documents. "Start Request" CTA navigates to submission form.
- **Feature 4: Status Tracking Dashboard (Simplified)**
Dashboard displays user's requests in table format. Columns: Request ID, Service Name, Submission Date, Status. Three status states only: Submitted, Processing, Completed. No status history or timeline view. Search by request ID only.
- **Feature 6: Document Upload (Basic)**
File upload via click-to-browse (no drag-and-drop). Supports PDF/JPEG/PNG up to 5MB. Client-side validation only (no virus scanning in MVP). Documents visible in request detail view with download links. No in-app document viewer.
- **Feature 3: Payment Gateway (Single Provider)**
Paymob integration only (no Fawry/multi-gateway). Credit/debit cards only (no wallets/bank transfer). Basic checkout flow: amount, card details, confirmation. Failed

payments show generic error and retry option. No refund automation (manual process).

- **Feature 5: Notifications (Email Only)**

Transactional emails only: request submitted, status changed to completed. No SMS, no push notifications. No notification preferences/settings. Uses existing YOUXEL SMTP infrastructure.

Out of MVP Scope:

- **Feature 2: Dynamic Forms**

Static forms deliver core functionality without form builder complexity. 90% of initial services use linear forms. Conditional logic needed for only 2 of 10 launch services. Development effort (3 person-months) too high for marginal UX improvement.

- **Advanced Search/Filtering (Catalog enhancement)**

With only 10 services at launch, category filters and faceted search are premature optimization. Basic text search sufficient for MVP catalog size.

- **Multi-Gateway Payment Support**

Paymob covers 85% of expected payment methods (credit/debit cards). Fawry/wallet integrations add 3 weeks dev time for 15% edge cases. Can manually process alternative payments backend.

- **Document Verification Workflow**

Staff review queue, approval/rejection UI, resubmission flows add complexity without validating core hypothesis: will users submit requests digitally? Can manually verify documents via email in MVP.

- **SMS/Push Notifications**

Email notifications meet compliance requirements (audit trail). SMS adds cost. Push requires mobile app (not in scope).

- **Status Timeline/History View**

Displaying granular status changes (5-7 stages) requires workflow engine and staff portal integration. MVP shows binary state: in-progress or done. Sufficient to answer "Is my request complete?"

- **Mobile Application**

Responsive web interface covers 80% of mobile use cases. Native app development (iOS + Android) requires 4 person-months + ongoing store maintenance.

Success Criteria:

The following KPIs will be monitored over 6 months post-launch to determine the success of the MVP:

1. Adoption rate: 50% of eligible requests submitted digitally (baseline 0%)
2. Median request processing time: Median 2.5 days (baseline 2 weeks)
3. Payment success rate: 92% first-attempt (industry benchmark)
4. Active users: 60% of user base
5. Completion rate: 85% of started requests finished
6. Support deflection: 40% reduction in "status inquiry" tickets

6. Conclusion

The E-services module represents more than just a functional upgrade; it is a foundational component of YOXEL's digital transformation. By transitioning from manual triage to automated, measurable workflows, the organization will gain unprecedented visibility into service bottlenecks and operational performance.

The RICE-based prioritization ensures that development resources are focused on the "vital few" features (Catalog, Tracking, and Payments) that resolve 80% of current user pain points. While the MVP is intentionally lean, it establishes the technical infrastructure (AWS ECS, PostgreSQL, and Modular API design) required to scale as user adoption grows.

Key Takeaways for Stakeholders:

- Value Realization: Success will be measured by adoption rates and the "support deflection" achieved by moving users away from phone/email.
- Risk Mitigation: Security and data privacy (GDPR) are integrated by design, not as afterthoughts.
- Scalability: The architecture is "microservices-ready," allowing the platform to evolve without requiring a total rewrite.

Upon successful validation of the MVP, the roadmap will expand to include sophisticated UX enhancements like dynamic forms and multi-gateway payment redundancies, further solidifying YOXEL's position as a digital-first leader.

Note: This proposal is built on strategic hypotheses that must be validated through data collection and user research before full-scale development.